Introduction - Programming

Programming: Everyday Decision-Making Algorithms

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Kühne Logistics University Hamburg - Winter 2025

About this Course

Teaching Team

Figure 1: Dr. Tobias Vlcek

Figure 2: Dr. Nils Roemer

About me

- Field: Optimizing and simulating complex systems
- Languages: of choice: Julia, Python and Rust
- Interest: Modelling, Simulations, Machine Learning
- Teaching: OR, Algorithms, and Programming
- Contact: vlcek@beyondsimulations.com

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> Tip
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> We really appreciate active participation and interaction!

Agenda

- About the course, team, and expectations
- How to learn programming (mindset and resources)
- Setting up Python with uv and VS Code

- Using notebooks with uv
- Q&A and next steps

Course Outline

- I: Optimal Stopping
- II: Explore & Exploit
- III: Caching
- IV: Scheduling
- V: Randomness
- VI: Computational Kindness

Participation

- Try actively participating in this course
- You will find it much (!) easier and more fun
- Lecture based on the book Algorithms to live by[1]
- Material and slides are hosted online: beyondsimulations.github.io/Programming-Everyday-Decisions

Teaching

- Lecture: Presentation and discussion of algorithms related to everyday decision-making
- Tutorial: Step-by-step assignments to be solved and discussed together in groups
- Difficulty: Strongly depends on your background and programming experience

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> Tip
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> No worries, we will help you out if you have any questions!

Passing the Course

- Pass/fail course without exams
- 75% attendance required for passing the course
- Hand in the assignments of at least two lectures
- Short presentation and discussion at the end
- You work together in groups of three students

Handing in Assignments

- Each student group submits one solution
- Provide us all working notebooks of the lecture
- Hand in is due at the beginning of the next lecture
- At least 50 % have to be correct to pass
- You have to pass at least twice

[1] Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.

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> Tip
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> This is just in order to provide you with working solutions after each deadline.

Learning Python

We will mostly not cover Python during the lectures!

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Question: Anybody know why?

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- In our experience, the best way to learn is by doing!
- Here, we will focus on decision-making algorithms
- You will learn Python by doing the tutorials

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> Tip
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> Don't worry, we will help you out if you have any questions!

Difficulty of the Course

- At first it might be a little bit overwhelming
- Programming is similar to learning a new language
- First, you have to get used to it and learn words
- Later, you'll be able to apply it and see results
- Important: Practice, practice, practice!

Goals of the Course

- Learn the basics of programming
- Learn about algorithmic thinking
- Be able to apply methods and concepts
- Solve practical problems with algorithms

> Tip
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> We are convinced that this course will be quite interesting and teach you more for your daily life than most other courses!

Why Python?

- Origins: Conceived in late 1980s as a teaching and scripting language
- Simple Syntax: Python's syntax is mostly straightforward and very easy to learn
- Versatility: Used in web development, data analysis, artificial intelligence, and more

- Community Support: A large community of users worldwide and extensive documentation

Help from AI

- You are allowed to use AI in the course, we use it as well (e.g., Claude, ChatGPT, LLama3 ...)
- These tools are great for learning Python!
- Can help you a lot to get started with programming

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> ⚠ Warning
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> But you should not simply use them to replace your learning.

How to learn programming

Our Recommendation

1. Be present: Attend the lecture and solve the tutorials
2. Put in some work: Repeat code and try to understand it
3. Do coding: Run code, play around, modify, and solve

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> Tip
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> Great resources to start are books and small challenges. You can find a list of book recommendations at the end of the lecture. Small challenges to solve can for example be found on Codewars.

Don't give up!

- Programming is problem solving, don't get frustrated!
- Expect to stretch your comfort zone

Setting up Python

Install VS Code

- Download and install from the website
- Built for Windows, Linux and Mac
- Install the Python and Jupyter extension
- Now you are ready to go!

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> Tip
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> Unsure on how to work with VS Code and notebooks? Ask us! We are happy to help you out!

What is an IDE?

- Integrated Development Environment `=` application
- It allows you to write, run and debug code scripts
- Other IDEs include for example:
  - PyCharm from JetBrains
  - Zed

Installation of Python with `uv`

- We will use `uv` to install and manage Python versions
- It works on Windows, Mac and Linux
- It helps us to manage packages and virtual environments
- Now, we all go here and install `uv` and Python

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> Tip
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> If the installation does not work, let us know!

Using Notebooks with uv

Quick Check

- Have you installed `uv` and initialized the project?
- Great! Before we continue, check the following:
  - [] You have a folder for the course
  - [] You have initialized `uv` with `uv init` inside the folder
  - [] You can see a file called `pyproject.toml` in the folder

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> Tip
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> Something not working yet? Ask us!

Using Notebooks

- Now we need to add a kernel to our project
- Run `uv add --dev ipykernel` from your terminal
- Now run `uv add jupyter` in the terminal
- This allows us to use `uv` Python in notebooks

- Done? Perfect. Now we can start!

Working with Notebooks

- Now you can download the files from the website
- Just click on one of the sessions and open it
- Select `Jupyter` on the right side
- Download and save the files to your course folder
- Open them and select "Open with Jupyter Notebook"

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> **Tip**
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> That was the hardest part today!

Any questions

so far?

After the break — Optimal Stopping

- The "Secretary Problem" and the 37% rule
- When to stop searching and make a decision
- How to translate the idea into code and experiments

> **Note**
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> That's it for our introduction!
> Let's have a short break and then continue with our first topic.

Literature

Interesting literature to start

- Christian, B., & Griffiths, T. (2016). Algorithms to live by: the computer science of human decisions. First international edition. New York, Henry Holt and Company.[2]
- Ferguson, T.S. (1989) 'Who solved the secretary problem?', Statistical Science, 4(3). doi:10.1214/ss/1177012493.

Books on Programming

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Here
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

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[2] The main inspiration for this lecture. Nils and I have read it and discussed it in depth, always wanting to translate it into a course.

> **Note**
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> Think Python is a great book to start with. It's available online for free. Schrödinger Programmiert Python is a great alternative for German students, as it is a very playful introduction to programming with lots of examples.

More Literature

For more interesting literature, take a look at the literature list of this course.